

HiMe: Real-Time Self-Hosted Personal Agent Platform for Health Management with Wearable Devices

Wei Liu^{*,*}, Siya Qi^{*,*}, Linhai Zhang^{*}, Lorraine Tudor Car^{*}, Yulan He^{*,*✉},

^{*}King’s College London, [✉]The Alan Turing Institute,
{wei.4.liu, yulan.he}@kcl.ac.uk

Abstract

Traditional approaches to wearable health signal analysis are constrained by rigid analytical frameworks and limited personalisation. The emergence of LLM agents creates a new opportunity for Personal Health Agentic Analysis, where health insights can be generated adaptively and in context. However, currently there is no open-source locally deployable platform capable of processing personal health data in real time while preserving privacy. We present HiMe, a locally deployable, privacy-first agent platform that is fully compatible with real-time health data ecosystems across a wide range of wearable devices. HiMe is guided by three design principles. The database is treated as a first-class component. Effectiveness and efficiency are jointly optimised to achieve a low-cost Pareto-optimal balance. Data are processed in real time while the user is modelled over the long term. Together, these principles make it practical for individuals to harness Personal Health Agents for continuous, personalised health management¹.

1 Introduction

Wearable devices have moved health monitoring out of the clinic and into everyday life. Modern smartwatches and smartphones continuously record signals such as heart rate, blood oxygen, sleep, and workouts into private on-device stores. Over time, these signals form a longitudinal record of an individual’s health. But making full use of this stream is hard. Conventional wearable analytics rely on fixed statistics that often ignore both signal dynamics and user preferences, while the long-term raw trace is too large to fit within the context window of a large language model (LLM) for intelligent analysis.

^{*}Equal contribution.

¹Available at <https://github.com/thinkwee/HiMe>

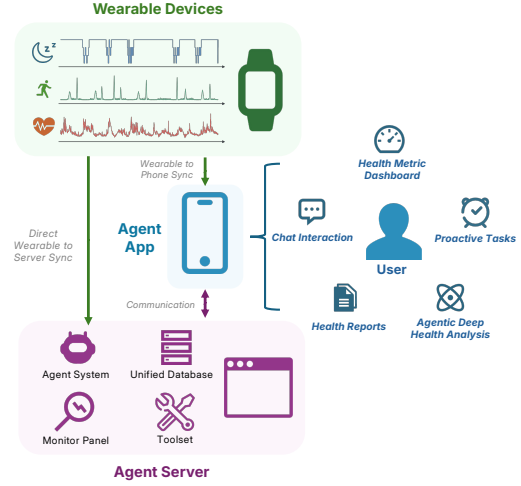


Figure 1: Overview of the HiMe system.

LLM agents that reason and act through tools offer a promising alternative (Yao et al., 2023). Such agents can access wearable streams through tools, decide what is worth analysing, and explain their findings in natural language. This promise has mainly been pursued in two settings: clinical multi-agent systems and personal coaching systems. But neither fully addresses the needs of individuals living with their own continuous health data. Clinical multi-agent systems reason over encounter-based records for clinician-facing use cases (Tang et al., 2024; Kim et al., 2024; Schmidgall et al., 2024), making the clinician, rather than the patient, the primary user. Personal coaching systems such as GPTCoach (Jörke et al., 2025) and MindScape (Nepal et al., 2024) are built around guided dialogue or journal prompts rather than a continuous stream that an agent can monitor and act upon proactively. General-purpose self-hosted assistants have also become increasingly widespread. OpenClaw (Steinberger, 2026), for example, popularised a local-first personal agent that reaches the user through messaging channels. Yet, such systems are primarily built for chat and scheduled tasks, not

for a high-rate physiological stream or months-long models of individuals. There is still no open-source platform that allows individuals to run, on their own hardware, an agent that ingests wearable data in real time and supports long-term personal health management.

We present HIME, the **Health Intelligence Management Engine**, a locally deployable and privacy-first agent platform that connects to real-time health data ecosystems across a wide range of wearable devices. HIME is built on three design principles for personal health management. **1)** The database is treated as a first-class component. This enables the agent to analyse wearable signals, personal records and its own memory jointly, while the checkable terminal state of the database provides a direct target for evaluating LLM agents on HIME’s health management tasks. **2)** Effectiveness and efficiency are optimised jointly, seeking a Pareto-optimal trade-off among quality, cost, and latency, allowing always-on agency to remain affordable on consumer hardware and local models. **3)** Real-time data processing is combined with long-term user modelling. This allows the system to capture both momentary anomaly and months-long trends, supporting timely intervention and longitudinal understanding. In personal health management, each user is effectively a continuous, heterogeneous data stream. Such personal data are an asset fundamentally different from the public text used to train large models. HIME is built to harness LLM agents for this stream, not only to satisfy a momentary request but to help users become a healthier self over time. The modular design also supports AI researchers requiring environments for evaluating and training health agents, and medical research applying large models to real-world wearable data.

2 Architecture

HIME is a self-hosted agent platform consisting of an agent server and agent apps on the smartphone and wearable devices (Figure 1). A data adapter decouples wearable formats and the signal collection pipeline from the agent, so HIME can ingest any wearable source and the evaluation in §3 replays public datasets through the same interface. Figure 2 shows the four layers of the platform.

Data layer. Everything the agent holds lives in a unified database with separate tables for memory, user profiles, and wearable signals, and the agent can create custom tables for recording anything it

needs. The agent can therefore analyse heterogeneous data jointly. When asked for a health check report, for example, the agent reads previously established baselines from its self-generated customisation tables and analysis skills from the skill table. It then queries the signal table over multiple steps, responds to the user, and writes the results to the report table. Moreover, it may also add a morning exercise proactive notification to the task table and update user preferences in the user table. Each agent holds its own permissions over the tables, ensuring that every operation remains safe. No workflow is hard-coded, and HIME only provides the tool set to the agent. HIME also encourages agents to create tables for their own use and provides schema design principles rather than fixed constraints. This design enables weaker LLMs to generate correctly formatted outputs, while allowing stronger LLMs to operate without being restricted by rigid schemas. By integrating heterogeneous data across different tables, agents obtain rich contextual information that grounds personalised and comprehensive responses.

Tool layer. With the database serving as a first-class component, the tool layer can be kept remarkably simple. In principle, only two types of tools are needed: read/write operations over the unified database, and a code execution tool. For finer-grained control, we implement scenario-specific variants, such as replying to users, modifying files, or generating dashboards, yet their underlying abstractions remain unchanged. The code execution tool maintains Jupyter-style state across calls (Kluyver et al., 2016), preserving the execution context and encouraging deeper multi-step exploration, as the agent never needs to rewrite a script from scratch. Tool visibility is further separated by agent role. To prevent tool-use hallucination, a fact verifier classifies every outbound message from its tool trace, checks each reported number against the corresponding query result, and stores a hashed evidence trail behind a show-evidence button for auditing.

Agent layer. HIME is designed for long-term interaction, so the interaction agent serves as the primary agent and owns all user-facing communication. At any step of its multi-step tool use, it can invoke specialised sub-agents as tools. These include an analysis agent for data analysis, a management agent for system operations such as memory updates and proactive task configuration, and

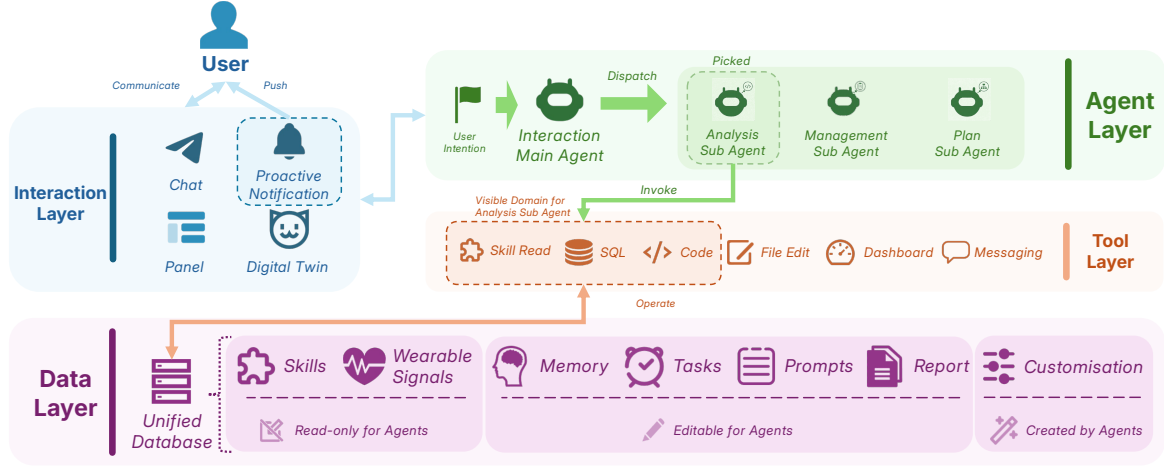


Figure 2: The four layers of the HiME framework.

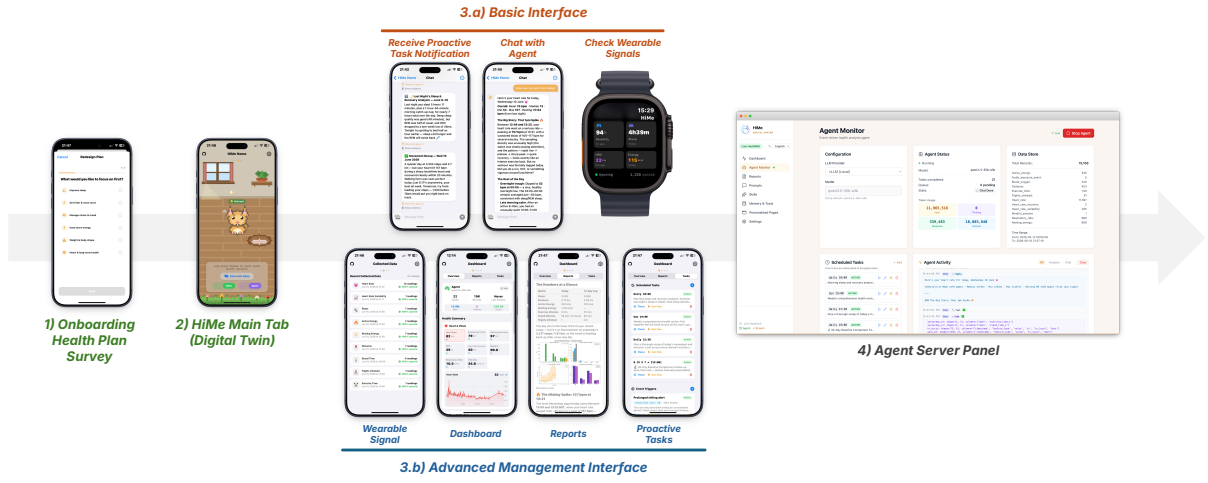


Figure 3: User-facing demonstration flow, from onboarding through a day of use across the in-app chat, wearable device status check, advanced management tabs and agent server panel for comprehensive monitoring.

a plan agent for personalised health plans. Agent inputs are assembled from modular prompt layers, including system instructions, roles, soul, responsibilities, experience and other reusable components. Composing these layers maximises prompt reuse and keeps the prompt prefix stable, improving key-value (KV) cache utilisation and reducing inference cost. The backend supports 15 LLM API providers, yet we strongly recommend serving open-source models locally through vLLM (Kwon et al., 2023) for a privacy-first deployment.

Interaction layer. Users reach the agent through the built-in chat of the HiME app or through integrated messaging platforms such as WeChat and Telegram. A gateway registry routes each reply back to the channel the message arrived on, falling back to another enabled channel when needed. As interactions accumulate, the agent keeps updating

its memory and the user preference profile, and configures proactive tasks autonomously for increasingly personalised health management. All prompts, memory files and proactive tasks stay fully editable in the agent server panel, either by hand or by instructing the agent. For a more engaging experience, a pixel-art cat acts as a digital twin, and its animations reflect the user’s current health status in real time. Beyond reactive replies, proactive tasks let the agent push notifications to the smartphone and the watch. These tasks cover both time-based schedules and event-based triggers, and they live in the task tables of the data layer under the agent’s own management. The agent authors these triggers itself. Figure 3 shows a session across these surfaces.²

²A screencast walkthrough accompanies the submission.

From principles to design. The three principles explain why the system takes this shape rather than the field’s defaults. **1)** Treating the database as a first-class component positions the agent as an interface over persistent state rather than a transient reasoning engine. Responsibilities then factor naturally into reads and writes over structured data, and the user model stays as agent-editable Markdown instead of an opaque vector store (Packer et al., 2023; Chhikara et al., 2025; Xu et al., 2025). Every action leaves a checkable terminal database state, so evaluation can be automatic and a new capability arrives as a new table rather than a new agent. **2)** Jointly optimising effectiveness and efficiency removes unnecessary computation while preserving capability. Lightweight statistical filters run before expensive inference, functionality decomposes into narrow roles with small prompts, each role routes to the cheapest adequate model, and a static prompt prefix improves KV-cache reuse. Principles rather than rigid workflow raise the floor for weak models without capping strong ones, keeping always-on agency feasible on consumer hardware at low cost. **3)** Pairing real-time processing with long-term modelling separates continuous sensing from episodic reasoning. Lightweight components monitor the stream at full resolution, and the LLM is invoked only upon meaningful triggers. A unified loop interleaves chat, events and scheduled tasks, and the prompt assembler injects the evolving long-term user model into every step, so transient anomalies and long-horizon trends are captured within the same framework. §3 analyses these choices.

3 Evaluation

HIME is a demonstration system rather than a benchmark, yet the first principle makes it measurable. Because every capability leaves a terminal database state, most actions can be scored by replaying a public wearable trace, letting the agent act and comparing the resulting state against a gold state computed independently from the same trace. We drive the deployed agent offline over the randomly sampled ten participants of each of five public wearable corpora, namely LifeSnaps (Yfantidou et al., 2022), PMData (Thambawita et al., 2020), MMASH (Rossi et al., 2020), CovIdentify (Cho et al., 2024) and GLOBEM (Xu et al., 2022). The sweep covers 22 backbones. Nineteen run locally and span the Qwen2.5 (Yang et al., 2024), Qwen3 (Yang et al., 2025), Qwen3.5 (Qwen

Team, 2026a), Qwen3.6 (Qwen Team, 2026b,c) and Gemma-4 (Google DeepMind, 2026) families from 1.5B to 35B parameters, along with nemotron-3-nano, granite-4.1-8b and glm-4.7-flash. GPT-5-mini, DeepSeek-V4-flash and Gemini-3.5-flash are hosted APIs, serving as a lightweight frontier reference. On-device execution is the right target for privacy and latency, so the question that matters is how far local backbones, small ones in particular, sit from that frontier. Each corpus enters through its own data adapter, making HIME an off-the-shelf, unified platform for wearable agent benchmarking and research.

3.1 Per-role agentic capability

We give every agent in HIME the same backbone and score it against a terminal-state oracle that needs no LLM judge, reported as pass@1 (Chen et al., 2021) over three runs. Each of the five role columns isolates one capability with its own closed-form check. **Route** scores the top-level dispatch. The interaction agent must hand a user message to exactly the sub-agents it needs, with every required role present and none spurious. **Analysis** answers a natural-language question over the replayed trace. It succeeds when the reported number $\hat{v}$ matches the value v of an independent reference query, $|\hat{v} - v| \leq \max(0.02 |v|, 0.5)$, and is backed by a real query or code run. **Memory** executes a write instruction. It succeeds when the resulting user profile and proactive task state match the request, with the protected file header intact and no collateral edits. **Plan** drives onboarding. It succeeds when the created scheduled tasks carry valid cron expressions, cover every stated health goal and ship with a published report. **Halluc.** captures the failure mode shared across these roles, the rate of asserting a number with no backing query, measured on the analysis tasks. Each role keeps its own oracle, so a failure is attributed to the role that caused it, a diagnosis that a single-role agent design cannot give.

Per-role results. Table 1 reports results for twelve representative backbones. Three findings stand out. First, local deployment rivals hosted frontier models. The strongest local backbones achieve an analysis score of 0.91 (Qwen3.6-27b and Qwen3.6-35b-a3b), matching or surpassing DeepSeek-V4-flash (0.85) and GPT-5-mini (0.76) while exhibiting no hallucinations. Only Gemini-3.5-flash clearly leads on analysis (0.94). Second,

Backbone	Route	Analysis	Memory	Plan	Halluc.↓	MT	S5	tok/sess
Gemini-3.5-flash [†]	0.98	0.94	0.89	0.98	0.01	0.98	0.37	42k
DeepSeek-V4-flash [†]	1.00	0.85	0.97	1.00	0.00	0.84	0.00	66k
GPT-5-mini [†]	0.88	0.76	0.99	0.88	0.00	0.63	0.00	206k
Qwen3.6-27b	1.00	0.91	1.00	0.99	0.00	0.76	0.20	253k
Qwen3.6-35b-a3b	1.00	0.91	0.98	0.84	0.00	0.96	0.26	169k
Qwen3-32b	1.00	0.68	0.81	0.80	0.00	0.69	0.00	178k
Gemma-4-26b-a4b	1.00	0.48	0.93	0.33	0.26	0.89	0.44	236k
Qwen3-8b	0.61	0.47	0.96	0.57	0.00	0.68	0.00	194k
Qwen3-4b	1.00	0.43	0.67	0.37	0.00	0.57	0.00	33k
Qwen3-1.7b	0.50	0.41	0.01	0.99	0.00	0.26	0.00	123k
Qwen3-30b-a3b	0.97	0.36	0.33	0.99	0.00	0.61	0.00	242k
Qwen2.5-1.5b	0.37	0.00	0.00	0.00	0.47	0.26	0.00	20k

Table 1: Per-role capability for twelve representative backbones, three hosted ([†]) and nine local, auto-scored on replayed public traces as pass@1 over three runs. MT is the mean per-turn score and S5 the strict all-five-turns rate of the composite rundown, Halluc. the unbacked-number rate, and tok/sess the mean tokens per five-turn session. The full 22-backbone sweep is in Table 5.

capability does not scale monotonically with model size. The Qwen3-30b-a3b mixture-of-experts model, despite having approximately 3B active parameters and 30B parameters in total, trails the smaller dense Qwen3-8b on analysis (0.36 versus 0.47) and performs poorly on memory (0.33 versus 0.96). Analysis is the most discriminative capability overall, ranging from 0.00 to 0.94, whereas routing performance saturates for most capable backbones. Third, small models lag substantially behind. The 1.5B model fails across all capabilities and exhibits the highest hallucination rate (0.47). Current small on-device models remain insufficient for personal health agent tasks.

Composite rundown. The per-role scores measure each capability in isolation, and isolation overstates what a user experiences. We therefore run a *composite rundown*, a five-turn session that interleaves several tasks. The session is one persistent chat ordered the way a real daily use unfolds. Turn one asks for a weekly check-in plan, scored on the created tasks covering both sleep and activity. Turn two asks the agent to recall what it just set up. Turn three asks for the average heart rate on the user’s own most active day and demands a number backed by a real query. Turn four states a durable preference that must land in memory. Turn five compounds into two, switches to metric units, and asks for the all-time peak heart rate, so both the grounded number and the written preference must be correct. With $a_{i,t}$ and $r_{i,t}$ marking turn t of session i passing its task and routing checks, over N sessions we report MT (Mean per Turn) score and

S5 (Strict all-5-turns pass) score,

$$\begin{aligned} \text{MT} &= \frac{1}{5N} \sum_{i,t} a_{i,t}, \\ \text{S5} &= \frac{1}{N} \sum_i \prod_t r_{i,t} a_{i,t}. \end{aligned} \tag{1}$$

Performance degrades over long interaction trajectories. Per-turn scores stay healthy, with Gemini-3.5-flash at 0.98 and Qwen3.6-35b-a3b at 0.96, yet a flawless session is rare because per-turn errors compound. The best S5 values are only 0.44 (Gemma-4-26b-a4b) and 0.37 (Gemini-3.5-flash), and most backbones sit at zero. Reliably chaining individually competent roles across a session, not single-turn skill, is therefore the open frontier for on-device agents.

Does decomposition help? Splitting the agent into separate roles is first an engineering choice. Each role owns a bounded slice of the task, so its behaviour is easy to manage and log, and a failure points to one role rather than to a tangle of tool calls inside a single persona. The ablation shows the split is also nearly free on the metric (Table 6). Most backbones that can run the multi-role protocol gain or break even, led by +0.65 for Qwen3-8b and +0.23 for Qwen3-32b. Only Qwen3.6-27b at −0.36 and Gemma-4-e4b at −0.71 pay a real cost. In real deployment, the trajectory grows long, and a monolithic persona must hold every role in one context, so the advantage of factored responsibilities widens.

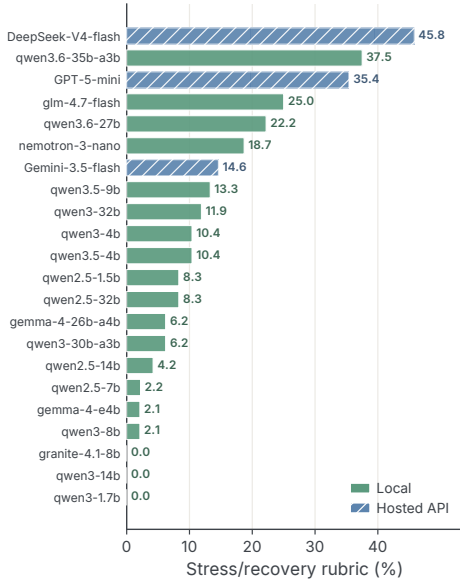


Figure 4: Cross-modal report understanding, the stress/recovery rubric score (0 to 100%) of a wearable-only summary judged against the participant’s held-out self-report. The breakdown is in Table 4.

3.2 Data-to-report evaluation

When asked for a personalised health report, the agent must read a participant’s multivariate history and commit to what it means, a cross-modal task from time series to a natural-language verdict that probes understanding rather than retrieval (Liu et al., 2026). A fixed LLM judge (Liu et al., 2023) scores the wearable-only summary against the participant’s held-out self-report with a rubric covering fatigue, readiness and stress. Figure 4 shows the task is hard. The best backbone reaches only 46% (DeepSeek-V4-flash). Gemini-3.5-flash leads single-turn analysis at 0.94 yet scores 14.6% here, while nemotron-3-nano at 0.85 analysis reaches 18.7%. Understanding and narrating subjective state is therefore a competence separate from retrieving numbers, and it is the one a personal agent adds on top of question answering.

3.3 Real-time triggering analysis

An always-on agent for healthcare must act unprompted, watching a continuous wearable stream without paying for an LLM call at every reading. HIME lets the agent compile a user’s intent into a trigger, a cheap statistical condition that runs continuously for free and wakes the expensive multi-turn analysis only when it fires (§2). To test whether this gating keeps detection intact while cutting cost, we compare the trigger against

Backbone	Polling recall	Trigger recall	Tokens saved
Qwen3-1.7b	0.00	0.29	121×
Qwen3-8b	0.69	0.89	76×
Gemma-4-26b-a4b	0.57	0.43	83×
Qwen3-30b-a3b	0.58	0.52	49×

Table 2: Detection on injected anomalies across PM-Data, LifeSnaps and MMASH, five participants per corpus and 47 events per backbone.

a polling baseline that runs the identical analysis on a fixed schedule. We test if the agent can discover injected anomalies in the health data stream (Appendix D gives the full protocol). Table 2 reports four backbones’ results on conditional recall given trigger activation. The agent-generated trigger analysis identifies more events on average compared with the polling baseline. The gains from agent-authored trigger-based analysis are more pronounced on smaller models. This is because fixed scheduled triggers cannot accurately pinpoint relevant data streams, causing smaller models to be overwhelmed by large volumes of normal signals and miss critical information. For larger models, in contrast, the approach achieves an order-of-magnitude reduction in token usage at the cost of only a modest decrease in recall.

3.4 Longitudinal Real-User Study[In Progress]

Whether HIME helps a person is a separate question that only real use can answer. The aim is not momentary satisfaction but a healthier self over time, so the study measures change over a sustained period rather than one session. We are running a two-month study in which eight non-technical participants use HIME continuously on their own devices and data. Table 3 reports, on a five-point scale, whether the system helped them better understand their health, manage it day to day, and improve a self-chosen health indicator, alongside a usability score (Brooke, 1996). Results will be reported for the camera-ready version.

4 Conclusion

We presented HIME, a self-hosted agent platform that transforms continuous wearable streams into long-term personal health management. Grounded in three design principles, its architecture is designed to be measurable, auditable, and extensible. Experiments across five public corpora and

Measure	Outcome
Better understands own health	◇
Manages own health day to day	◇
Improvement on a chosen indicator	◇
System Usability Scale (0 to 100)	◇

Table 3: Longitudinal study with eight participants over two months of continuous use. The first three rows are self-reported on a five-point scale. **This section is still [in progress].**

22 backbones show that strong local models can already rival hosted APIs, suggesting that privacy-preserving personal health agents are increasingly practical. At the same time, important challenges remain, particularly in multi-turn reliability and faithful data-to-report narration. Despite these limitations, our longitudinal study provides initial evidence that personal health agents can support improved long-term health outcomes. By releasing HIME, we hope to facilitate future research by individuals, agent developers, and medical researchers toward trustworthy and personalised health agents.

References

- John Brooke. 1996. SUS: A ‘quick and dirty’ usability scale. In Patrick W. Jordan, B. Thomas, Ian Lyall McClelland, and Bernard Weerdmeester, editors, *Usability Evaluation in Industry*, pages 189–194. Taylor & Francis.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, and 1 others. 2021. [Evaluating large language models trained on code](#). *arXiv preprint arXiv:2107.03374*.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. [Mem0: Building production-ready AI agents with scalable long-term memory](#). *arXiv preprint arXiv:2504.19413*.
- Peter Cho, Md Mobashir Hasan Shandhi, Ali Roghanizad, and Jessilyn Dunn. 2024. [CovIdentify dataset](#). PhysioNet. Version 1.0.0.
- Google DeepMind. 2026. Gemma 4: Byte for byte, the most capable open models. <https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/>.
- Matthew Jörke, Shardul Sapkota, Lyndsea Warkenthien, Niklas Vainio, Paul Schmiedmayer, Emma Brunskill, and James A. Landay. 2025. [GPTCoach: Towards LLM-based physical activity coaching](#). In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery.
- Yubin Kim, Chanwoo Park, Hyewon Jeong, Yik Siu Chan, Xuhai Xu, Daniel McDuff, Hyeonhoon Lee, Marzyeh Ghassemi, Cynthia Breazeal, and Hae Won Park. 2024. [MDAgents: An adaptive collaboration of LLMs for medical decision-making](#). In *Advances in Neural Information Processing Systems*.
- Thomas Kluyver, Benjamin Ragan-Kelley, Fernando Pérez, Brian Granger, Matthias Bussonnier, Jonathan Frederic, Kyle Kelley, Jessica Hamrick, Jason Grout, Sylvain Corlay, Paul Ivanov, Damián Avila, Safia Abdalla, Carol Willing, and Jupyter Development Team. 2016. [Jupyter notebooks – a publishing format for reproducible computational workflows](#). In *Positioning and Power in Academic Publishing: Players, Agents and Agendas*, pages 87–90. IOS Press.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. [Efficient memory management for large language model serving with PagedAttention](#). In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, pages 611–626. ACM.
- Wei Liu, Peijie Yu, Michele Orini, Yali Du, and Yulan He. 2026. [Hunt instead of wait: Evaluating deep data research on large language models](#). *arXiv preprint arXiv:2602.02039*.
- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. [G-Eval: NLG evaluation using GPT-4 with better human alignment](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2511–2522. Association for Computational Linguistics.
- Subigya Nepal, Arvind Pillai, William Campbell, Talie Massachi, Michael V. Heinz, Ashmita Kunwar, Eunsoo Soul Choi, Xuhai Xu, Joanna Kuc, Jeremy F. Huckins, Jason Holden, Sarah M. Preum, Colin Depp, Nicholas C. Jacobson, Mary P. Czerwinski, Eric L. Granholm, and Andrew T. Campbell. 2024. [Mind-Scape study: Integrating LLM and behavioral sensing for personalized AI-driven journaling experiences](#). *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 8(4).
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2023. [MemGPT: Towards LLMs as operating systems](#). *arXiv preprint arXiv:2310.08560*.
- Qwen Team. 2026a. Qwen3.5: Towards native multi-modal agents. <https://qwen.ai/blog?id=qwen3.5>.
- Qwen Team. 2026b. Qwen3.6-27B: Flagship-level coding in a 27B dense model. <https://qwen.ai/blog?id=qwen3.6-27b>.

- Qwen Team. 2026c. Qwen3.6-35B-A3B: Agentic coding power, now open to all. <https://qwen.ai/blog?id=qwen3.6-35b-a3b>.
- Alessio Rossi, Eleonora Da Pozzo, Dario Menicagli, Chiara Tremolanti, Corrado Priami, Alina Sirbu, David Clifton, Claudia Martini, and David Morelli. 2020. [Multilevel monitoring of activity and sleep in healthy people](#). PhysioNet. Version 1.0.0.
- Samuel Schmidgall, Rojin Ziaei, Carl Harris, Eduardo Reis, Jeffrey Jopling, and Michael Moor. 2024. [AgentClinic: A multimodal agent benchmark to evaluate AI in simulated clinical environments](#). *arXiv preprint arXiv:2405.07960*.
- Peter Steinberger. 2026. OpenClaw: Your own personal AI assistant. Any OS. Any platform. <https://github.com/openclaw/openclaw>. Open-source software project, first released as Clawdbot in November 2025. Transfer to the non-profit OpenClaw Foundation announced February 2026.
- Xiangru Tang, Anni Zou, Zhuosheng Zhang, Ziming Li, Yilun Zhao, Xingyao Zhang, Arman Cohan, and Mark Gerstein. 2024. [MedAgents: Large language models as collaborators for zero-shot medical reasoning](#). In *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics.
- Vajira Thambawita, Steven Alexander Hicks, Hanna Borgli, Håkon Kvale Stensland, Debesh Jha, Martin Kristoffer Svensen, Svein-Arne Pettersen, Dag Johansen, Håvard Dagenborg Johansen, Susann Dahl Pettersen, Simon Nordvang, Sigurd Pedersen, Anders Gjerdrum, Tor-Morten Grønli, Per Morten Fredriksen, Ragnhild Eg, Kjeld Hansen, Siri Fagernes, Christine Claudi, and 7 others. 2020. [PMDData: A sports logging dataset](#). In *Proceedings of the 11th ACM Multimedia Systems Conference (MMSys)*, pages 231–236.
- Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025. [A-MEM: Agentic memory for LLM agents](#). *arXiv preprint arXiv:2502.12110*.
- Xuhai Xu, Han Zhang, Yasaman Sefidgar, Yiyi Ren, Xin Liu, Woosuk Seo, Jennifer Brown, Kevin Kuehn, Mike Merrill, Paula Nurius, Shwetak Patel, Tim Althoff, Margaret Morris, Eve Riskin, Jennifer Mankoff, and Anind Dey. 2022. [GLOBEM dataset: Multi-year datasets for longitudinal human behavior modeling generalization](#). In *Advances in Neural Information Processing Systems (NeurIPS): Datasets and Benchmarks Track*.
- An Yang, Anfeng Li, Baosong Yang, and 1 others. 2025. [Qwen3 technical report](#). *arXiv preprint arXiv:2505.09388*.
- An Yang, Baosong Yang, Beichen Zhang, and 1 others. 2024. [Qwen2.5 technical report](#). *arXiv preprint arXiv:2412.15115*.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. [ReAct: Synergizing reasoning and acting in language models](#). In *The Eleventh International Conference on Learning Representations (ICLR)*.
- Sofia Yfantidou, Christina Karagianni, Stefanos Efstathiou, Athena Vakali, Joao Palotti, Dimitrios Pan-teleimon Giakatos, Thomas Marchioro, Andrei Kazlouski, Elena Ferrari, and Šarūnas Girdzijauskas. 2022. [LifeSnaps, a 4-month multi-modal dataset capturing unobtrusive snapshots of our lives in the wild](#). *Scientific Data*, 9(1):663.

A Report Rubric Breakdown

This appendix reports the complete result tables behind §3. All numbers come from the same replayed-trace protocol and terminal-state oracles as the main text. Table 4 breaks the data-to-report rubric of §3.2 into its three dimensions on 16 PMData participants, a larger PMData-only sample than the ten-per-corpus sweep used for the per-role evaluation. The dimensions are not equally hard. Fatigue peaks at 62.5% and stress at 68.8%, yet readiness never passes 31.2%, so anticipating how ready a participant will feel is the hardest judgement. The ranking also reshuffles the per-role order. DeepSeek-V4-flash leads with the most balanced profile, Qwen3.6-35b-a3b is the best local backbone on the strength of its stress score, and three backbones, namely granite-4.1-8b, Qwen3-14b and Qwen3-1.7b, score zero on every dimension.

Backbone	Overall%	Fatigue%	Readi.%	Stress%
DeepSeek-V4-flash [†]	45.8	62.5	31.2	43.8
Qwen3.6-35b-a3b	37.5	25.0	18.8	68.8
GPT-5-mini [†]	35.4	37.5	18.8	50.0
glm-4.7-flash	25.0	37.5	25.0	12.5
Qwen3.6-27b	22.2	20.0	20.0	26.7
nemotron-3-nano	18.7	37.5	6.2	12.5
Gemini-3.5-flash [†]	14.6	6.2	18.8	18.8
Qwen3.5-9b	13.3	20.0	0.0	20.0
Qwen3-32b	11.9	7.1	14.3	14.3
Qwen3-4b	10.4	6.2	18.8	6.2
Qwen3.5-4b	10.4	12.5	6.2	12.5
Qwen2.5-1.5b	8.3	6.2	12.5	6.2
Qwen2.5-32b	8.3	18.8	0.0	6.2
Gemma-4-26b-a4b	6.2	6.2	6.2	6.2
Qwen3-30b-a3b	6.2	18.8	0.0	0.0
Qwen2.5-14b	4.2	6.2	6.2	0.0
Qwen2.5-7b	2.2	0.0	0.0	6.7
Gemma-4-e4b	2.1	0.0	0.0	6.2
Qwen3-8b	2.1	0.0	0.0	6.2
granite-4.1-8b	0.0	0.0	0.0	0.0
Qwen3-14b	0.0	0.0	0.0	0.0
Qwen3-1.7b	0.0	0.0	0.0	0.0

Table 4: Full cross-modal report results, the stress/recovery rubric percentage and the per-dimension hit rate for fatigue, readiness and stress, judged against held-out self-reports on 16 PMData participants. [†] hosted API.

B Full Per-Role Sweep

Table 5 extends Table 1 of §3.1 to all 22 backbones, sorted by the analysis score. The full sweep sharpens the main findings. The newest local generations close most of the gap to hosted APIs, and Qwen3.5-9b stands out as the cheapest competent configuration, reaching 0.87 analysis with perfect memory and plan scores at only 44k tokens per session. Single-turn skill does not guarantee session-level reliability, since nemotron-3-nano pairs 0.85

analysis with a collapsed multi-turn score of 0.02. Routing saturates at or near 1.00 for over half the field yet falls to 0.37 at the 1.5B floor. Hallucination concentrates in the older Qwen2.5 family and in Gemma-4-26b-a4b. Token spend per session spans an order of magnitude and does not buy quality, as Qwen3-30b-a3b spends 242k tokens for 0.36 analysis while Qwen3.5-9b spends 44k for 0.87.

C Decomposition Ablation

Table 6 reports the single-turn composed task of §3.1 for the 13 local backbones of the ablation run, decomposed against monolithic. The other nine backbones of the sweep postdate this run. The full list bears out the main text. Six backbones gain from the split, led by Qwen3-8b at +0.65 and Qwen3-32b at +0.23, and five more sit within 0.09 of break-even. Only Qwen3.6-27b at −0.36 and Gemma-4-e4b at −0.71 pay a real cost, while the two smallest backbones fail under either protocol. Factoring the agent into roles therefore costs little even on a single turn, the setting least favourable to it.

D Trigger Detection Protocol

This appendix details the detection experiment of §3.3. We compare two ways of invoking the *same* agent over the same stream, differing only in when the LLM is woken. The **polling** baseline runs a full multi-turn agentic analysis on a fixed five-minute schedule, the exhaustive upper bound that inspects everything. The **trigger** arm runs the identical analysis only when an agent-authored condition fires. That analysis is the deployed engine, which investigates the recent window with SQL and code tools across several turns before writing a report. The trigger rules are generated automatically by the evaluated backbone. Each model compiles one natural-language monitoring request, through the same path the deployed agent uses, into a handful of rules, each a statistical condition over one signal paired with a natural-language analysis goal.

We replay PMData, LifeSnaps and MMASH, the three corpora with an intraday heart-rate stream, through the same data adapter as §3 under a frozen clock and fixed ingestion batches, so every trailing-window query and cooldown matches the live system. Because no corpus labels real anomalies, we inject diverse events of controlled magnitude relative to each user’s own recent baseline, namely heart-rate spikes, bradycardic drops, sus-

Backbone	Rt	An	Mem	Pl	H↓	MT	S5	tok
Gemini-3.5-flash [†]	0.98	0.94	0.89	0.98	0.01	0.98	0.37	42k
Qwen3.6-27b	1.00	0.91	1.00	0.99	0.00	0.76	0.20	253k
Qwen3.6-35b-a3b	1.00	0.91	0.98	0.84	0.00	0.96	0.26	169k
Qwen3.5-9b	1.00	0.87	1.00	1.00	0.00	0.91	0.01	44k
nemotron-3-nano	0.99	0.85	0.93	0.66	0.00	0.02	0.00	52k
DeepSeek-V4-flash [†]	1.00	0.85	0.97	1.00	0.00	0.84	0.00	66k
Qwen3.5-4b	1.00	0.77	0.84	0.18	0.00	0.66	0.00	50k
GPT-5-mini [†]	0.88	0.76	0.99	0.88	0.00	0.63	0.00	206k
Qwen2.5-32b	1.00	0.72	0.66	0.07	0.00	0.74	0.22	159k
Qwen3-32b	1.00	0.68	0.81	0.80	0.00	0.69	0.00	178k
granite-4.1-8b	0.60	0.64	0.93	0.80	0.00	0.00	0.00	34k
Qwen3-14b	1.00	0.52	0.79	0.00	0.00	0.44	0.00	162k
Gemma-4-26b-a4b	1.00	0.48	0.93	0.33	0.26	0.89	0.44	236k
Qwen3-8b	0.61	0.47	0.96	0.57	0.00	0.68	0.00	194k
Qwen3-4b	1.00	0.43	0.67	0.37	0.00	0.57	0.00	33k
Qwen3-1.7b	0.50	0.41	0.01	0.99	0.00	0.26	0.00	123k
Qwen2.5-7b	0.60	0.38	0.66	0.91	0.12	0.31	0.00	38k
Qwen3-30b-a3b	0.97	0.36	0.33	0.99	0.00	0.61	0.00	242k
glm-4.7-flash	1.00	0.27	0.83	0.92	0.00	0.00	0.00	33k
Qwen2.5-14b	0.74	0.15	0.63	0.22	0.22	0.62	0.00	102k
Gemma-4-e4b	1.00	0.02	0.67	0.00	0.00	0.37	0.00	144k
Qwen2.5-1.5b	0.37	0.00	0.00	0.00	0.47	0.26	0.00	20k

Table 5: Full per-role capability over 22 backbones (pass@1, three runs), sorted by analysis. Rt route, An analysis, Mem memory, Pl plan, H hallucination, MT multi-turn per-turn score, S5 strict all-five-turns, tok mean tokens per five-turn session. [†] hosted API.

Backbone	Dec	Mono	Δ
Qwen3-8b	0.98	0.33	+0.65
Qwen3-32b	0.52	0.29	+0.23
Qwen2.5-7b	0.21	0.09	+0.12
Qwen3-14b	0.77	0.66	+0.11
Qwen3-30b-a3b	0.29	0.22	+0.07
Gemma-4-26b-a4b	0.69	0.63	+0.06
Qwen3-1.7b	0.00	0.00	0.00
Qwen2.5-1.5b	0.00	0.00	0.00
Qwen3.6-35b-a3b	0.90	0.91	-0.01
Qwen2.5-32b	0.07	0.11	-0.04
Qwen2.5-14b	0.80	0.89	-0.09
Qwen3.6-27b	0.63	0.99	-0.36
Gemma-4-e4b	0.01	0.72	-0.71

Table 6: Decomposition ablation on the single-turn composed task, decomposed (Dec, three roles) versus monolithic (Mono, one persona, all tools), for the 13 local backbones in the ablation run, sorted by Δ (decomposed minus monolithic).

tained tachycardia, oxygen-desaturation dips and depressed heart-rate variability, as the detection ground truth: five participants per corpus and 47 events per backbone. A fixed LLM judge (Liu et al., 2023) then scores whether the woken analysis identifies each event. We report recall on the events whose gate fires, which isolates analysis quality from how completely the agent authored its gate, and the token saving in Table 2 is the ratio of the trigger arm’s few daily analyses to the polling baseline’s fixed schedule.

E Human Evaluation Survey Design

[In progress.] This appendix will document the survey instrument behind the longitudinal user study of §3.4. It will set out how the questionnaire is designed, the items each participant answers at the recurring check-ins, the response scales and the administration schedule across the two-month deployment, together with the wording of every question. The instrument is still being finalised and will be reported here in full in a later revision.